Constrained Lifecycle Optimization of Low-Carbon Structural Concrete Under 11-Year Natural Carbonation Kinetics and Compressive Strength Demands

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***Abstract*:** Supplementary cementitious materials reduce emissions but deplete alkaline buffering against carbonation. Using an 11-year dataset of 45 mixtures, we formulate a constrained lifecycle optimization framework. With IS 10262 strength margins, discrete screening selects 50% slag Mix C24 at $\geq$ 30 mm cover under EN 206; IS 456 Severe excludes C24, while C40/50 selects C23. At 25 mm cover, kinetic uncertainty increases carbon by 16.5%. Across 18 scenarios, continuous optimization predicts 4.6%–31.2% carbon savings (1.1%–29.0% with PCE). At 30 mm cover, net global warming potential (GWP) is 133.4 kg CO_2e/m^3, saving 14.1 kg/m^3 over C24 (8.8–9.9 kg/m^3 with PCE). Uptake offsets 3.8% to 5.7% of emissions. No original mixture used PCE; chemical compaction to 2% air remains uncalibrated.

***Index Terms*:** Low-carbon concrete, natural carbonation kinetics, lifecycle optimization, supplementary cementitious materials, service-life durability, Pareto frontier.

I. Introduction

A. The Decarbonization Mandate for Structural Concrete

Global warming potential (GWP) expresses greenhouse-gas impacts as carbon dioxide equivalent (CO_2e).

Annual concrete production exceeds 30 billion tonnes. Clinker calcination ($CaCO_3 \rightarrow CaO + CO_2$) and fuel combustion contribute 7%–8% of global CO_2 [1], [2]. Standard life-cycle inventories report benchmark OPC emissions of 0.832–0.840 kg CO_2e/kg [1], [3]. Reported Indian kiln factors span 0.893–0.940 kg CO_2e/kg (893–940 kg/t) in plant audits and cradle-to-gate LCAs [4].

Supplementary cementitious materials (SCMs), including blast-furnace slag (slag), siliceous fly ash (FA), limestone filler (LS), and volcanic pozzolan (PO), replace clinker with industrial by-products [1], [5]. Clinker replacement also creates a durability trade-off. Hydration of cement produces Portlandite ($Ca(OH)_2$), which maintains pore solution pH > 12.5 and passivates steel reinforcement through a microscopic oxide film (γ-Fe_2O_3). Atmospheric CO_2 diffuses into unsaturated pores and neutralizes this alkaline reserve:

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O \tag{1}$$

This reaction lowers pore solution pH below 9.0 [6]. When the front reaches the reinforcement depth (d_{cover}), steel depassivates, initiating active corrosion and cover spalling [7], [8]. SCMs contain less reactive CaO and consume Portlandite pozzolanically. High clinker replacement therefore depletes the alkaline buffering capacity [9].

B. Research Gaps and Problem Formulation

Accelerated carbonation tests (1%–4% CO_2) alter pore microstructure relative to natural exposure [10], [11]. Prior optimizations treat strength or carbon in isolation, omitting reactive transport, entrapped air, and codified bounds.

This study uses the $\approx$ 11-year Chennai dataset of 45 mixes [11], [12] to examine how clinker replacement can reduce lifecycle carbon while satisfying strength and carbonation-cover constraints. The framework separates field measurements, calibrated models, and literature-based scenarios into three evidence tiers. It combines a cross-validated, air-normalized Bolomey strength model with sheltered and unsheltered carbonation kinetics, including $+1\sigma$ margins, to screen measured mixtures and optimize continuous compositions across 18 scenarios. Closed-form durability bounds and shadow prices then explain how cover and composition limits control the selected mixtures. The notation is summarized in Table I.

II. Decade-Long Natural Weathering Dataset and Transport Physics

A. Experimental Campaign and Field Exposure

The experimental campaign comprises 45 concrete mixtures formulated with Portland cement, fly ash (15%–50%), slag (30%–50%), limestone (10%–20%), pozzolan (15%–30%), and ternary blends (30%–40%), exposed under sheltered and unsheltered conditions in Chennai, India [11], [12]. Water-to-binder ratios span $w/b \in [0.45, 0.60]$ with binder $B \in [220, 400]$ kg/m^3 (Mix C1 is the sole mix at $B = 220$ kg/m^3, 0% SCM; SCM blends require $B \geq 250$ kg/m^3).

Concrete prisms were stored for 7–104 days (mean: 39.8 days; median: 28 days), with no statistically significant effect of storage duration observed ($p > 0.50$; Appendix A). The dataset contains 353 carbonation depth measurements across six testing periods spanning 1.0–11.5 years, including 167 sheltered and 186 unsheltered measurements. The final testing period had a mean exposure age of 10.75 years. Fresh concrete air content ranged from 0.9% to 6.6%, averaging 2.5%. The study evaluates three target compressive strengths (30, 35, and

TABLE I
MATHEMATICAL NOTATION, PARAMETERS, AND PHYSICAL UNITS

Symbol	Physical Meaning / Definition	Unit
$\mathbf{x}$	Decision vector $[w/b, B, x_{\mathrm{FA}}, x_{\mathrm{SG}}, x_{\mathrm{LS}}, x_{\mathrm{PO}}]^T$	—
w/b, B/W	Water-to-binder mass ratio and reciprocal B/W	—
B	Total cementitious binder content	$\mathrm{kg/m^3}$
x_j, p_j	Mass fraction (x_j) and percentage ($p_j = 100x_j$) of SCM j	—, %
Air%	Entrapped batch fresh air volume percentage	%
$f_{c,28}, f_{\mathrm{target}}$	28-day cylinder compressive strength and structural demand	MPa
$\Delta f_{\mathrm{margin}}$	Codified strength screening buffer (+8.25 MPa, IS 10262)	MPa
$x_c(t), k, d_{50}$	Carbonation depth, rate ($x_c = k\sqrt{t}$), and 50-yr depth	$\mathrm{mm}, \mathrm{mm/yr^{0.5}}$
$d_{\mathrm{cover}}, d_{50,\mathrm{allow}}$	Reinforcement cover depth and allowable nominal depth	mm
$D_{\mathrm{eff}}, a, C_{\mathrm{ext}}$	Effective CO_2 diffusivity, buffer capacity, ambient CO_2	$\mathrm{m^2/s}, \mathrm{mol/m^3}$
S_w, ϕ_p, ϕ_g	Pore saturation ($0 \le S_w \le 1$), total and gas porosity	—
$GWP_{\mathrm{net}}, GWP_{\mathrm{A1-A3}}$	Net lifecycle GWP and cradle-to-gate embodied carbon	$\mathrm{kg\ CO_2e/m^3}$
$\mathrm{CO}^{\mathrm{B1}}_{2,\mathrm{up}}$	Use-phase 50-year stoichiometric carbonation uptake	$\mathrm{kg\ CO_2/m^3}$
$EF_i, EF_{\mathrm{batching}}$	Material emission factor (i) and batching footprint	$\mathrm{kg/kg}, \mathrm{kg/m^3}$
μ_k	Karush-Kuhn-Tucker dual multiplier (shadow price) of constraint k	unit^{-1}

40 MPa) and three concrete cover depths (25, 30, and 35 mm) under both exposure conditions.

B. Earth System Reactive-Transport Governing Equations

Concrete is an unsaturated geochemical porous medium undergoing reactive-diffusive exchange with atmospheric CO_2. The 1D conservation law for gaseous CO_2 concentration $C(x,t)$ $[\mathrm{mol/m^3}]$ in unsaturated pores is:

$$\frac{\partial(\phi_g C)}{\partial t} = \frac{\partial}{\partial x}\left(D_{\mathrm{eff}}(S_w, \phi_p, T)\frac{\partial C}{\partial x}\right) - r_{\mathrm{carb}} \tag{2}$$

where $\phi_g = \phi_p(1 - S_w)$ is gas-filled porosity, ϕ_p is total connected porosity, $S_w \in [0,1]$ is liquid saturation, and r_{carb} is the volumetric mineralization sink ($\mathrm{mol\ CO_2/(m^3 \cdot s)}$).

Under the Millington-Quirk tortuosity-porosity formulation:

$$D_{\mathrm{eff}}(S_w) = D_0 \cdot \frac{\phi_g^{10/3}}{\phi_p^2} = D_0 \cdot \phi_p^{4/3}(1 - S_w)^{10/3} \tag{3}$$

where $D_0 \approx 1.6 \times 10^{-5}\ \mathrm{m^2/s}$ is molecular CO_2 diffusivity in air. In pore water, aqueous diffusivity decreases to $D_{\mathrm{aq}} \approx 1.8 \times 10^{-9}\ \mathrm{m^2/s} \approx 10^{-4} D_0$. In outdoor exposure (XC4), periodic monsoon rain induces capillary saturation ($S_w \to 1$), reducing $D_{\mathrm{eff}} \to 0$ and providing a plausible physical explanation for the 26%–30% empirical carbonation rate attenuation observed outdoors relative to sheltered exposure (XC3).

The assumptions are instantaneous reaction-front propagation ($r_{\mathrm{carb}} \to \infty$ at $x = x_c(t)$, M1), quasi-steady pore diffusion ($\partial^2 C/\partial x^2 \approx 0$ in $0 \le x \le x_c$, M2), and boundary values $C(0,t) = C_{\mathrm{ext}}$, $C(x_c,t) = 0$ (M3).

Integrating Laplace's equation under M3 gives a linear concentration gradient $\partial C/\partial x = -C_{\mathrm{ext}}/x_c(t)$. Applying the Stefan mass-conservation condition across the moving interface yields:

$$a \cdot \frac{\mathrm{d}x_c}{\mathrm{d}t} = -D_{\mathrm{eff}} \left.\frac{\partial C}{\partial x}\right|_{x=x_c^-} = D_{\mathrm{eff}} \frac{C_{\mathrm{ext}}}{x_c(t)} \tag{4}$$

where $a = [\mathrm{Ca(OH)_2}]_0 + \beta[\mathrm{C\text{-}S\text{-}H}]_0$ is total volumetric alkaline buffering capacity ($\mathrm{mol\ CO_2/m^3}$). Separating variables and integrating from $x_c(0) = 0$:

$$\begin{aligned} \int_0^{x_c(t)} x'\,\mathrm{d}x' &= \frac{D_{\mathrm{eff}} C_{\mathrm{ext}}}{a}\int_0^t \mathrm{d}t' \\ \implies \frac{1}{2}x_c(t)^2 &= \frac{D_{\mathrm{eff}} C_{\mathrm{ext}}}{a} t \end{aligned} \tag{5}$$

Taking the square root recovers the exact Fickian rate law:

$$x_c(t) = k\sqrt{t}, \quad \text{with } k = \sqrt{\frac{2 D_{\mathrm{eff}} C_{\mathrm{ext}}}{a}} \tag{6}$$

Replacing clinker with SCMs reduces a, amplifying k. Slag hydration produces secondary C-S-H that retains alkaline reserve and refines pore structure, whereas fly ash pozzolanic reactions consume Portlandite while generating slow early microstructure, explaining why $\gamma_{\mathrm{FA}}/\gamma_{\mathrm{SG}} = 2.17\times$.

When continued hydration causes time-decaying diffusivity $D(t) = D_0(t_0/t)^\alpha$ ($\alpha \in [0,1)$), the moving boundary obeys anomalous sub-diffusive power-law penetration:

$$\begin{aligned} x_c(t) &= \left[\frac{2 D_0 C_{\mathrm{ext}} t_0^\alpha}{a(1-\alpha)}\right]^{1/2} t^{(1-\alpha)/2} \\ &= k'\, t^n, \quad n = \frac{1-\alpha}{2} \end{aligned} \tag{7}$$

In sheltered exposure, negligible post-curing hydration aging yields $\alpha \approx 0 \implies n \approx 0.50$ (empirical pooled sheltered fit $n = 0.509$, 95% CI $[0.458, 0.560]$). Because $n = 0.509$ is slightly above the theoretical bound ($n = (1-\alpha)/2 \le 0.50$), it is interpreted as $\alpha \approx 0$ within fitting error. For outdoor monsoon exposure, rain-induced late hydration yields $\alpha \approx 0.10 \implies n = 0.452 < 0.50$ (95% CI $[0.406, 0.498]$).

C. Authoritative Three-Tier Evidence Demarcation

The analysis separates its inputs and outputs into three evidence tiers, summarized in Table II. Tier 1 contains measured data, including carbonation depths, cylinder strengths, fresh air contents, and Chennai climate records. Tier 2 contains strength and kinetic models calibrated against the experimental cohort; the calibrated kinetic laws have $R^2 > 0.86$. These fitted relationships support interpolation within the observed mixture domain but do not make every optimized composition an experimentally tested mixture.

Tier 3 contains literature-based scenarios and extrapolations, including projected service-life depths and lifecycle carbon estimates. External inputs include EN 16757 uptake [13] and EN 206 binder limits [14]. This distinction is retained when interpreting the optimization results: predicted performance at the design horizon and beyond the tested composition domain remains conditional on the adopted models and assumptions.

III. PREDICTIVE MODELING: STRENGTH, KINETICS, AND STOICHIOMETRY

A. Gel-Space Void Ratio and Compressive Strength Formulation

Compressive strength in hardened cementitious systems is dictated by paste porosity and solid-to-void ratio. Following

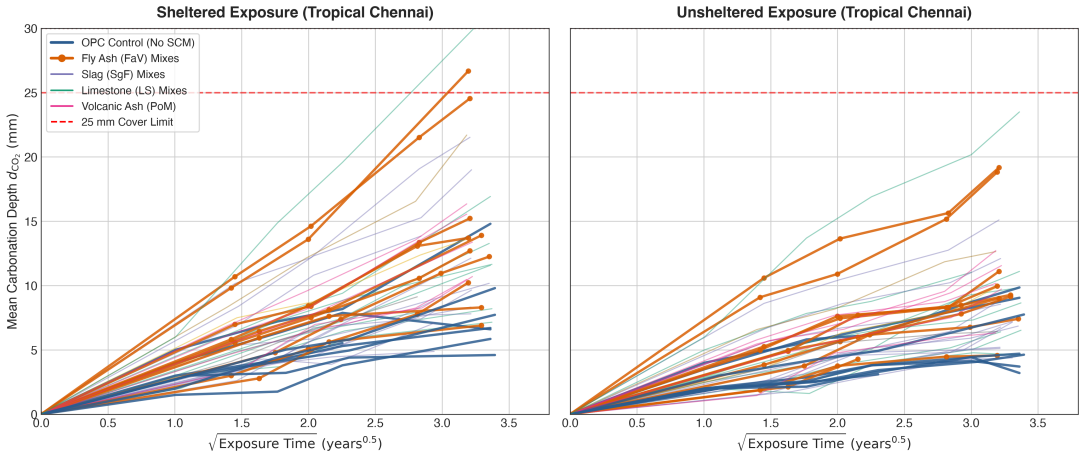


Fig. 1. Longitudinal natural carbonation kinetics (d vs. $\sqrt{t}$) across 11 years (final epoch 10.1–11.5 yr, mean 10.75 yr) of natural field exposure in Chennai, India: (a) Sheltered exposure (indoor/covered, EN 206 class XC3), showing accelerated penetration driven by dry convective CO_2 diffusion; (b) Unsheltered outdoor exposure (rain-exposed, EN 206 class XC4), illustrating substantial carbonation attenuation (26%–30% lower rates), consistent with physical pore-blocking from periodic monsoon wetting and pore condensation.

TABLE II
AUTHORITATIVE SCIENTIFIC DEMARCATION OF EXPERIMENTAL AND MODEL VARIABLES ACROSS THREE EVIDENCE TIERS

Variable / Quantity	Tier	Empirical Grounding	Model / External Basis	Defensible Scientific Claim
Carbonation Depth (x_c)	**Tier 1**	353 phenolphthalein tests, 6 epochs (1.0–11.5 yr, final epoch mean 10.75 yr)	Indicator spraying; corner-filtered spatial average [11]	Direct field measurements over $\approx$ 11 years in tropical Chennai.
28-Day Strength ($f_{c,28}$)	**Tier 1**	45 cylinder tests per EN 12390-3	Laboratory testing per standard protocol	Direct cylinder strength; characteristic targets $f_{ck,\mathrm{cyl}} = 30, 35, 40$ MPa map to cube M40, M45, M50 per EN 206 / IS 456.
Fresh Air Content (Air%)	**Tier 1**	Batch air volume (0.9%–6.6%, mean 2.5%, median 1.8%)	Direct pressure gauge measurement	Entrapped air (0.9%–6.6%, Mix C24 was 1.6%) dictates compressive strength.
Carbonation Rate (k)	**Tier 2**	OLS origin regression across 45 mixes over $\approx$ 11 years	$x_c = k\sqrt{t}$ (curing omitted as $p > 0.50$)	Mean sheltered $k = 3.73$, unsheltered 2.75 mm/yr$^{0.5}$. Uncertainty margin $+15\%$ matches $+1\sigma$ on $\ln k$.
Strength Model (f_c^{model})	**Tier 2**	Multivariable OLS on B/W, Air%, and SCM fractions	Bolomey solid-to-void ratio with air void volume correction	Sample RMSE 3.78 MPa, $s_e = 4.11$ MPa, LOO-CV 4.38 MPa. Buffer $+8.25$ MPa is IS 10262 codified margin ($+4.4$ MPa sensitivity).
Binder Domain (B)	**Tier 1/3**	Batch records: SCM mixes exist only for $B \in [250, 400]$ kg/m^3	Structural minimum binder rules	Mix C1 is sole mix at $B = 220$ kg/m^3 (0% SCM). Structural baselines enforce $B \geq 280$ (XC3) and $B \geq 300$ (XC4).
50-Year Depth (d_{50})	**Tier 3**	Field monitoring terminates at 10.1–11.5 yr (mean 10.75 yr)	Fickian diffusion projection ($t = 50$ years)	Extrapolates ≈ 4.3–$5.0\times$ beyond field observations; subject to $+15\%$ kinetic uncertainty.
CO_2 **Mass Uptake**	**Tier 3**	Measured depth x_c and binder content B	Stoichiometric mineralization chemistry	Cohort uptake offsets 3.8%–5.7% (baseline 4.7%, optimal 4.7%–7.7%, benchmark 6.4%–6.8%, 9.1–9.8 kg/m^3).
Embodied Carbon (GWP)	**Tier 3**	Mix constituent masses in kg/m^3	Standard LCI emission factors (ICE v3.0 [3], Scrivener et al. [1])	Cradle-to-gate A1–A3 minus Module B1 uptake. Baseline 0.840 kg/kg OPC; Indian plants (0.893–0.940 [4]) scale savings.

Powers' gel-space ratio $\mathcal{X} = V_{\mathrm{gel}}/(V_{\mathrm{gel}} + V_{\mathrm{cap}} + V_{\mathrm{air}})$ and Feret's law:

$$f_c = K_F \left(\frac{V_B}{V_B + V_W + V_{\mathrm{air}}} \right)^2 \approx K_B \left(\frac{B/\rho_B}{B/\rho_B + W/\rho_W + V_{\mathrm{air}}} - c_0 \right) \quad (8)$$

Linearizing Eq. 8 with respect to $B/W = 1/(w/b)$ and entrapped air volume Air% yields the multivariable Bolomey formulation:

$$f_{c,28}(\mathbf{x}) = \beta_0 + \beta_{B/W} \left(\frac{B}{W} \right) + \beta_{\mathrm{air}}(\mathrm{Air}\%) + \beta_{\mathrm{FA}} p_{\mathrm{FA}} + \beta_{\mathrm{SG}} p_{\mathrm{SG}} + \beta_{\mathrm{LS}} p_{\mathrm{LS}} + \beta_{\mathrm{PO}} p_{\mathrm{PO}} \quad (9)$$

where p_j are SCM mass percentages (0%–50%). Uncorrected regression of f_c against B/W shows breakdown ($R^2 = 0.071$, Fig. 2a), as low-w/b batches had high air (3.9%–6.6%, mean 5.1%; w/b–air $r = -0.585$, B/W–air $r = 0.642$; VIFs ≤ 2.09, ruling out collinearity). Normalizing for batch air volume restores the linear Bolomey trend ($R^2 = 0.734$, adjusted $R^2 = 0.692$, Fig. 2b). The empirical air penalty ($\beta_{\mathrm{air}} = -4.354$ MPa/%, $p < 10^{-6}$) reflects compaction void spaces. In practice, achieving $\leq 2.0\%$ air at $w/b = 0.45$ (126 L/m^3 water) relies on polycarboxylate ether superplasticizers with defoamers (Appendix A).

B. Model Calibration, Leverage, and +8.25 MPa Design Buffer

Calibration of Eq. 9 across all $N = 45$ concrete mixtures gives the parameter estimates in Table III. Slag exhibits latent hydraulic reactivity with a modest slope ($\beta_{\mathrm{SG}} = -0.098$ MPa/%, $p = 0.107$, 95% CI: $[-0.218, +0.022]$), whereas fly ash incurs a severe early penalty ($\beta_{\mathrm{FA}} = -0.421$ MPa/%, $p < 10^{-5}$).

Table III distinguishes sample RMSE (df = 45), residual error (df = 38), and LOO-CV error computed via diagonal hat-matrix leverage h_{ii}. For any out-of-sample design mixture $\mathbf{x}_0$,

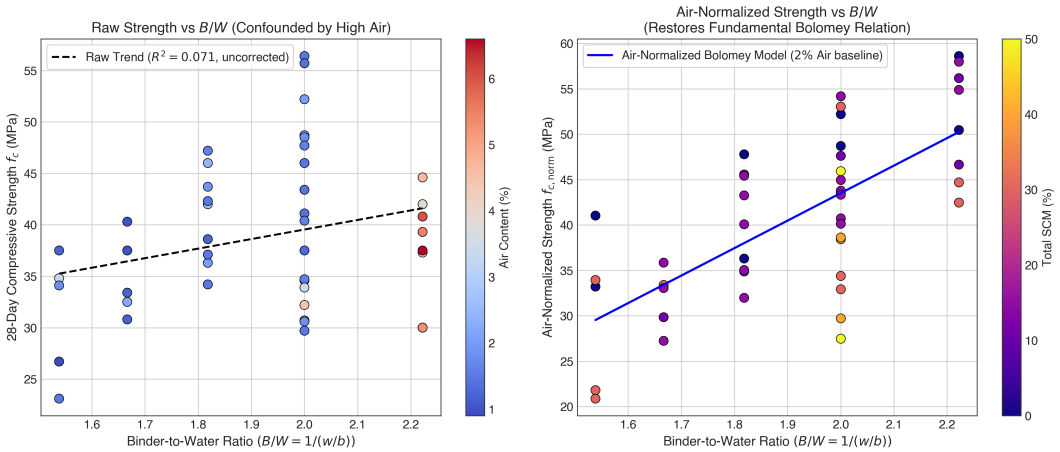


Fig. 2. Compressive strength modeling, Bolomey relation verification, and air void normalization across all $N = 45$ concrete mixtures: (a) Apparent breakdown of classical strength vs. B/W relationship prior to air void correction ($R^2 = 0.071$ against B/W, or $R^2 = 0.095$ against w/b); (b) Air-normalized compressive strength ($f_{c,\text{norm}}$ normalized to 2.0% baseline air content) vs. B/W, restoring the fundamental linear Bolomey trend across supplementary cementitious material replacement levels (0%–50%).

TABLE III
AIR-NORMALIZED BOLOMEY COMPRESSIVE STRENGTH MODEL PARAMETERS AND ERROR METRICS ($N = 45$)

Model Parameter / Metric	Estimate / Value
Intercept (β_0)	-3.698 MPa
Bolomey Slope ($\beta_{B/W}$)	$+30.959$ MPa
Air Void Penalty (β_{air})	-4.354 MPa/%
Fly Ash Slope (β_{FA})	-0.421 MPa/%
Slag Slope (β_{SG})	-0.098 MPa/%
Limestone Slope (β_{LS})	-0.467 MPa/%
Pozzolan Slope (β_{PO})	-0.589 MPa/%
Coefficient of Determination (R^2 / Adj R^2)	0.734 / 0.692
Sample RMSE ($N = 45$)	**3.78** MPa
Residual Standard Error (s_e, df = 38)	**4.11** MPa
Leave-One-Out CV Error (RMSE_{CV})	**4.38** MPa ($R^2_{\text{CV}} = 0.644$)
Codified Screening Buffer (Δf_{margin}, IS 10262)	$+$**8.25** MPa

prediction variance accounts for estimation uncertainty through leverage $h_{00} = \mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$:

$$\begin{aligned} s^2_{\text{pred}}(\mathbf{x}_0) &= s_e^2\left(1 + \mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0\right) \\ &= s_e^2(1 + h_{00}) \end{aligned} \tag{10}$$

For boundary corners ($h_{00} \approx 0.250 > \bar{h} = 0.156$), $s_{\text{pred}} = s_e\sqrt{1 + h_{00}} = 4.60$ MPa (exact 4.6007 MPa). We enforce a **codified design screening buffer**:

$$\begin{aligned} f^{\text{model}}_{c,28}(\mathbf{x}) &\geq f_{\text{target}} + \Delta f_{\text{margin}} \\ &= f_{\text{target}} + 8.25 \text{ MPa} \end{aligned} \tag{11}$$

from IS 10262:2019 Table 2 ($s = 5.0$ MPa for M30–M60; $+1.65s = +8.25$ MPa exceeds Table 1's $X = 6.5$ MPa). Sensitivities retain $s = 4.0$ MPa (+6.6 MPa) and LOO-CV +4.4 MPa. Target demands $f_{\text{target}} = 30, 35, 40$ MPa represent cylinder characteristic strengths $f_{ck,\text{cyl}}$ (tested per EN 12390-3), mapping conservatively to cube grades M40, M45, and M50 per EN 206 and IS 456 ($f_{ck,\text{cube}} \approx 1.25 f_{ck,\text{cyl}}$). This yields continuous design thresholds of 38.25 MPa for C30/37, 43.25 MPa for C35/45, and 48.25 MPa for C40/50. For discrete mixtures, nominal screening uses measured $f_{c,\text{meas}} \geq f_{\text{target}}$; buffered screening enforces $f_{c,\text{meas}} \geq f_{\text{target}} + 8.25$ MPa.

C. Carbonation Kinetic Modeling and SCM Contrast Ratio

Natural carbonation follows Fickian diffusion: $x_c(t) = k\sqrt{t}$, where k is regressed through the origin across the six epochs: $k = \sum(x_{c,i}\sqrt{t_i})/\sum t_i$ (Fig. 1). Multivariable log-linear models capture binder chemistry, w/b, and microclimate ($q_j = \text{SCM}_j\%/10.0$):

$$\begin{aligned} \ln k_{\text{sh}} = &-1.2939 + 4.0681(w/b) + 0.2525q_{\text{FA}} \\ &+ 0.1161q_{\text{SG}} + 0.3089q_{\text{LS}} + 0.2555q_{\text{PO}} \end{aligned} \tag{12}$$

($R^2 = 0.876$, $F = 54.92$, $p < 10^{-15}$ for Sheltered XC3), and:

$$\begin{aligned} \ln k_{\text{un}} = &-1.8130 + 4.4797(w/b) + 0.2482q_{\text{FA}} \\ &+ 0.0986q_{\text{SG}} + 0.3135q_{\text{LS}} + 0.2543q_{\text{PO}} \end{aligned} \tag{13}$$

($R^2 = 0.863$, $F = 49.15$, $p < 10^{-15}$ for Unsheltered XC4). In log space, fly ash exhibits a marginal kinetic slope $\gamma_{\text{FA}}/\gamma_{\text{SG}} = 0.2525/0.1161 = \mathbf{2.17}\times$ steeper than slag: +10% fly ash accelerates carbonation by +28.7% ($\exp(0.2525) = 1.287$), whereas +10% slag increases k by only +12.3% ($\exp(0.1161) = 1.123$).

For sheltered XC3, in-sample RMSE $= 0.1411$, $s_e = 0.1516$ (df = 39), and $\text{RMSE}_{\text{CV}} = 0.1663$; for unsheltered XC4, RMSE $= 0.1552$, $s_e = 0.1667$, and $\text{RMSE}_{\text{CV}} = 0.1765$. As an operational benchmark, we enforce a $+1\sigma$ margin ($\exp(0.1411) = 1.1515 \approx 1.15$):

$$k_{\text{design}} = \exp(\ln k_{\text{model}} + 0.1411) \approx 1.15\, k_{\text{model}} \tag{14}$$

imposing a +15% **kinetic uncertainty margin** (one-sided $+1\sigma$ band covering $\approx 84\%$ of normal cases). For 50-year depth $d_{50} = k_{\text{model}}\sqrt{50}$, enforcing $d_{50,\text{design}} \leq d_{\text{cover}}$ requires:

$$\begin{aligned} &d_{50,\text{design}} = 1.15\, k_{\text{model}}\sqrt{50} \leq d_{\text{cover}} \\ \iff\ &d_{50} \leq \frac{d_{\text{cover}}}{1.15} = d_{50,\text{allow}} = 0.8696 \cdot d_{\text{cover}} \end{aligned} \tag{15}$$

yielding $d_{50,\text{allow}} = 21.74$ mm for 25 mm cover, 26.09 mm for 30 mm, and 30.43 mm for 35 mm cover.

TABLE IV
STOICHIOMETRIC PARAMETERS FOR BOUNDING CO_2 UPTAKE SCENARIOS

Binder Constituent	Param.	Low	Mid (Base)	High
OPC Clinker	$\mathrm{CaO}_{\mathrm{OPC}}$	0.62	**0.65**	0.67
	$\alpha_{c,\mathrm{OPC}}$	0.60	**0.70**	0.80
Slag (GGBS)	$\mathrm{CaO}_{\mathrm{SG}}$	0.36	**0.40**	0.44
	$\alpha_{c,\mathrm{SG}}$	0.45	**0.60**	0.70
Fly Ash (FA)	$\mathrm{CaO}_{\mathrm{FA}}$	0.02	**0.05**	0.08
	$\alpha_{c,\mathrm{FA}}$	0.35	**0.50**	0.55
Limestone (LS)	$\mathrm{CaO}_{\mathrm{LS}}$	0.00	**0.00**	0.00
	$\alpha_{c,\mathrm{LS}}$	0.00	**0.00**	0.00
Pozzolan (PO)	$\mathrm{CaO}_{\mathrm{PO}}$	0.03	**0.05**	0.08
	$\alpha_{c,\mathrm{PO}}$	0.35	**0.50**	0.55

D. Stoichiometric CO_2 Mineralization and Chebyshev's Sum Inequality

Atmospheric carbonation mineralizes calcium hydroxide and C-S-H:

$$\mathrm{Ca(OH)_2} + \mathrm{CO_2} \rightarrow \mathrm{CaCO_3} + \mathrm{H_2O} \tag{16}$$

$$\text{C-S-H} + 2\,\mathrm{CO_2} \rightarrow \mathrm{SiO_2} \cdot m\mathrm{H_2O} + 2\,\mathrm{CaCO_3} \tag{17}$$

Following standardized uptake accounting (EN 16757, Module B1; Appendix A) and Lagerblad [15], volumetric use-phase CO_2 uptake (kg CO_2/m^3) over 50 years is:

$$\mathrm{CO}^{\mathrm{B1}}_{2,\mathrm{uptake}}(50\mathrm{y}) = x_c(50\mathrm{y}) \cdot 10^{-3} \cdot B \cdot \left(\frac{S}{V}\right) \cdot \Phi_{\mathrm{chem}} \left(\frac{44.01}{56.08}\right) \tag{18}$$

where $S/V = 5.0\ \mathrm{m^2/m^3}$ (standard 200 mm structural slab exposed on one face) and $\Phi_{\mathrm{chem}} = \sum_i f_i \cdot \mathrm{CaO}_i \cdot \alpha_{c,i}$ (Table IV). For the C35/45 benchmark defined in Section V-D, evaluating Eq. 18 using constituent products ($\Phi_{\mathrm{chem}} = 0.296$) gives 9.75 kg CO_2/m^3 (6.8% offset of 142.5 kg/m^3 embodied GWP, Net GWP = 132.8 kg/m^3). Decoupled evaluation ($\overline{\mathrm{CaO}} \cdot \bar{\alpha}_c = 0.441 \times 0.626 = 0.276$) gives 9.10 kg CO_2/m^3 (6.4% offset, Net GWP = 133.4 kg/m^3).

Constituent calcium contents and carbonation reactivities have the same descending order across the primary constituents:

$$\mathrm{CaO_{OPC}} > \mathrm{CaO_{SG}} > \mathrm{CaO_{FA}} \geq \mathrm{CaO_{LS}} \tag{19}$$

$$\alpha_{c,\mathrm{OPC}} > \alpha_{c,\mathrm{SG}} > \alpha_{c,\mathrm{FA}} \geq \alpha_{c,\mathrm{LS}} \tag{20}$$

Chebyshev's algebraic sum inequality guarantees that for any convex blending fractions ($f_i \geq 0, \sum f_i = 1$):

$$\sum_{i=1}^{m} f_i \mathrm{CaO}_i \alpha_{c,i} \geq \left(\sum_{i=1}^{m} f_i \mathrm{CaO}_i\right)\left(\sum_{i=1}^{m} f_i \alpha_{c,i}\right) = \overline{\mathrm{CaO}} \cdot \bar{\alpha}_c \tag{21}$$

Thus, decoupled evaluation is a guaranteed lower bound within the stoichiometric model, underestimating uptake by ≈ 0.65 kg/m^3 and yielding conservative Net GWP. Cohort uptake offsets 3.8%–5.7% of initial GWP (mean 4.7%, 11.4 kg/m^3); for optimal mixes, uptake offsets 4.7%–7.7% (6.4%–6.8% benchmark).

IV. CONTINUOUS LIFECYCLE OPTIMIZATION FORMULATION

A. Modeling Assumptions

The optimization model uses six explicit assumptions:

- **A1 1D Fickian Diffusion**: Sharp mineralization front $x_c(t) = k\sqrt{t}$ [6].
- **A2 Decoupled Kinetics**: Rate $\ln k$ is log-linear in w/b and SCM fractions ($\alpha = 0$).
- **A3 Bolomey Strength**: 28-d strength is linear in B/W, Air%, and SCM substitution.
- **A4 Linear LCI**: Cradle-to-gate carbon (A1–A3) is strictly linear and additive [3].
- **A5 Stoichiometric Uptake**: B1 uptake is lower-bounded via Chebyshev's inequality.
- **A6 Compaction**: Nominal air $V_{\mathrm{air}} = 2.0\%$ via superplasticizer (Appendix A).

B. Decision Vector, Objective, and Numbered Constraints

The continuous mix optimization problem is defined by the decision vector

$$\mathbf{x} = [w/b,\ B,\ x_{\mathrm{FA}},\ x_{\mathrm{SG}},\ x_{\mathrm{LS}},\ x_{\mathrm{PO}}]^T \in \mathbb{R}^6$$

where $x_j \in [0,1]$ represents the mass fraction of cementitious constituent j. Clinker mass fraction is determined by mass conservation $x_{\mathrm{OPC}} = 1.0 - \sum_{j \in \mathrm{SCM}} x_j$. Constituent masses per cubic meter ($m_i = B \cdot x_i$) and batch water ($m_{\mathrm{w}} = B \cdot (w/b)$) satisfy absolute volume conservation:

$$m_{\mathrm{agg}} = \left(1.0 - \sum_i \frac{m_i}{\rho_i} - \frac{m_{\mathrm{w}}}{\rho_{\mathrm{w}}} - V_{\mathrm{air}}\right) \cdot \rho_{\mathrm{agg}} \tag{22}$$

using the following constituent material densities (all in kg/m^3): $\rho_{\mathrm{OPC}} = 3150$, $\rho_{\mathrm{FA}} = 2300$, $\rho_{\mathrm{SG}} = 2900$, $\rho_{\mathrm{LS}} = 2700$, $\rho_{\mathrm{PO}} = 2500$, $\rho_{\mathrm{agg}} = 2650$, and $\rho_{\mathrm{w}} = 1000$.

Initial cradle-to-gate embodied carbon (Modules A1–A3) is:

$$GWP_{\mathrm{A1-A3}}(\mathbf{x}) = \sum_i m_i EF_i + m_{\mathrm{agg}} EF_{\mathrm{agg}} + m_{\mathrm{w}} EF_{\mathrm{w}} + EF_{\mathrm{batching}} \tag{23}$$

where standard emission factors are $EF_{\mathrm{OPC}} = 0.840$, $EF_{\mathrm{FA}} = 0.015$, $EF_{\mathrm{SG}} = 0.060$, $EF_{\mathrm{LS}} = 0.025$, $EF_{\mathrm{PO}} = 0.030$, $EF_{\mathrm{agg}} = 0.006$, $EF_{\mathrm{w}} = 0.0005$ kg CO_2e/kg [3], and $EF_{\mathrm{batching}} = 7.5$ kg CO_2e/m^3.

The 50-year use-phase stoichiometric CO_2 uptake (Module B1) is:

$$\mathrm{CO}^{\mathrm{B1}}_{2,\mathrm{uptake}}(\mathbf{x}, 50\mathrm{y}) = x_c(50\mathrm{y}, \mathbf{x}) \cdot 10^{-3} \cdot B \cdot \left(\frac{S}{V}\right) \cdot (\overline{\mathrm{CaO}} \cdot \bar{\alpha}_c) \left(\frac{44.01}{56.08}\right) \tag{24}$$

where $S/V = 5.0\ \mathrm{m^2/m^3}$. Net lifecycle GWP subtracts estimated use-phase CO_2 uptake from production emissions:

$$\min_{\mathbf{x}}\ \mathcal{J}(\mathbf{x}) = GWP_{\mathrm{A1-A3}}(\mathbf{x}) - \mathrm{CO}^{\mathrm{B1}}_{2,\mathrm{uptake}}(\mathbf{x}, 50\mathrm{y}) \tag{25}$$

TABLE V
OPTIMAL MIX AND NET CARBON COMPARISON ACROSS 18 DESIGN SCENARIOS

A. Sheltered (XC3)

		Nominal		Buffered		Continuous	
Grade	**Cover**	**Mix**	**GWP**	**Mix**	**GWP**	**Base**	**+15%**
C30/37	25	**C39**	169.2	**C23**	197.1	**135.7**	**137.5**
C30/37	30	**C24**	147.6	**C24**	147.6	**133.4**	**135.2**
C30/37	35	**C24**	147.6	**C24**	147.6	**131.6**	**133.3**
C35/45	25	**C39**	169.2	**C23**	197.1	**135.7**	**137.5**
C35/45	30	**C24**	147.6	**C24**	147.6	**133.4**	**135.2**
C35/45	35	**C24**	147.6	**C24**	147.6	**133.2**	**133.3**
C40/50	25	**C39**	169.2	**C23**	197.1	**135.8**	**137.5**
C40/50	30	**C24**	147.6	**C23**	197.1	**135.8**	**135.8**
C40/50	35	**C24**	147.6	**C23**	197.1	**135.8**	**135.8**

B. Unsheltered (XC4)

		Nominal		Buffered		Continuous	
Grade	**Cover**	**Mix**	**GWP**	**Mix**	**GWP**	**Base**	**+15%**
C30/37	25	**C24**	151.2	**C24**	151.2	**142.2**	**143.7**
C30/37	30	**C24**	151.2	**C24**	151.2	**142.0**	**142.0**
C30/37	35	**C24**	151.2	**C24**	151.2	**142.0**	**142.0**
C35/45	25	**C24**	151.2	**C24**	151.2	**144.2**	**144.2**
C35/45	30	**C24**	151.2	**C24**	151.2	**144.2**	**144.2**
C35/45	35	**C24**	151.2	**C24**	151.2	**144.2**	**144.2**
C40/50	25	**C24**	151.2	**C23**	200.1	**146.9**	**146.9**
C40/50	30	**C24**	151.2	**C23**	200.1	**146.9**	**146.9**
C40/50	35	**C24**	151.2	**C23**	200.1	**146.9**	**146.9**

Cover: mm; GWP: kg CO_2e/m^3. Discrete nominal: measured strength, no margin; buffered: +8.25 MPa and +15% kinetics. Continuous: +8.25 MPa strength with baseline or +15% kinetics. Compositions, screening counts, conditions, and qualifications: Appendix B.

subject to the following numbered constraints:

$$g_{\text{str}}(\mathbf{x}) = (f_{\text{target}} + \Delta f_{\text{margin}}) - f_{c,28}^{\text{model}}(\mathbf{x}) \leq 0 \tag{26}$$

$$g_{\text{dur}}(\mathbf{x}) = \gamma_{\text{dur}}\, x_c(50\text{y}, \mathbf{x}) - d_{\text{cover}} \leq 0 \tag{27}$$

$$g_{\text{scm}}(\mathbf{x}) = \sum_{j \in \text{SCM}} x_j - 0.50 \leq 0 \tag{28}$$

$$g_{\text{FA}}(\mathbf{x}) = x_{\text{FA}} - 0.35 \leq 0 \tag{29}$$

$$g_{\text{bnd}}(\mathbf{x}) : w/b \leq w/b_{\max}, \quad B \geq B_{\min},\ x_j \geq 0 \tag{30}$$

where $\gamma_{\text{dur}} = 1.0$ (nominal) or 1.15 (uncertain); strength margins follow Section III-B. Bound limits enforce structural baselines ($B \geq 280$ kg/m^3 for XC3; $B \geq 300$ kg/m^3 for XC4; $w/b \leq 0.55$ for XC3; $w/b \leq 0.50$ for XC4).

The Lagrangian associated with Eq. 25 is $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = \mathcal{J}(\mathbf{x}) + \sum_k \mu_k g_k(\mathbf{x})$. At local optimum $\mathbf{x}^*$, the first-order Karush-Kuhn-Tucker (KKT) conditions require:

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\mu}^*) = \mathbf{0} \tag{31}$$

$$\mu_k^* g_k(\mathbf{x}^*) = 0, \quad \mu_k^* \geq 0, \quad g_k(\mathbf{x}^*) \leq 0 \quad \forall k \tag{32}$$

By the envelope theorem, the dual multiplier $\mu_k^* = -\partial \mathcal{J}^* / \partial b_k$ represents the marginal shadow price of constraint k. In Section V, we link μ_{dur}^* directly to the carbon dividend of structural cover.

Structural Baselines vs. Extrapolation: Codified binder and FA baselines enforce minimum cover ($d_{50} = d_{\text{cover}}$). Section VI discusses paste starvation [16] and unvalidated SCM extrapolation at $B = 220$ kg/m^3.

V. LIFECYCLE OPTIMIZATION RESULTS AND PERFORMANCE ANALYSIS

A. Performance Across the Eighteen Design Scenarios

Table V summarizes optimal concrete mixtures across all nine structural design cases evaluated under sheltered and unsheltered microclimates (18 total scenarios). Each case combines a target strength with a cover depth, allowing the effects of strength demand, cover, and exposure to be compared within the same framework. The table reports discrete screening alongside continuous optimization, with nominal and buffered results shown separately. Discrete winners are selected from measured mixtures, whereas continuous solutions are model-predicted compositions subject to the specified strength, durability, and binder constraints. Appendix B provides the corresponding compositions and screening counts. The following subsections examine the constraints that change the selected mixtures and the carbon savings associated with continuous design.

B. Discrete Screening and the 25 mm Uncertainty Penalty

Table V gives the discrete winners; their controlling boundaries are:

1) **The 25 mm Sheltered Dilemma (Mix C39 vs. C23)**: Nominal C39 clears C40/50 by only 0.4 MPa. Codified margins disqualify it for $\geq$ C35/45, while $+15\%$ kinetics ($1.15 \times 22.6 = 26.0 > 25$ mm) eliminates it across all grades. C23 is the unique buffered survivor, imposing a $+16.5\%$ carbon penalty.
2) **Dominance of Mix C24 at ≥ 30 mm Cover**: C24 dominates C30/37 and C35/45 under nominal, model-buffered and IS 10262 $s = 4$ screens ($d_{50} = 25.0$ mm, 28.8 mm buffered), saving 25.1% (49.5 kg/m^3) over C23. EN 206 permits it; IS 456 Severe excludes it. The main $s = 5.0$ MPa margin eliminates C24 for C40/50 and promotes C23.
3) **Unsheltered Microclimate Performance**: Rain pore-blocking retards C24 carbonation to $d_{50} = 17.52$ mm. C24 wins C30/37 and C35/45 across covers; C23 wins C40/50 under $s = 5.0$ MPa. The $w/b \leq 0.50$ pool spans 24/26 nominally and 5/26 for C40/50 under joint buffering.

C. Continuous Optimization: Binder Reduction vs. SCM Blending

Continuous optimization reduces binder content from $B = 310$ to 280 kg/m^3 while lowering w/b from 0.50 to 0.45 and adjusting the SCM blend. The decomposition assigns 70.8%–87.1% of the savings to binder reduction, or approximately 79% overall, and 12.9%–29.2% to SCM blending. The bilinear interaction is not statistically significant ($p > 0.80$). Tables V and VII give all compositions and carbon values; the additional case-specific predictions are:

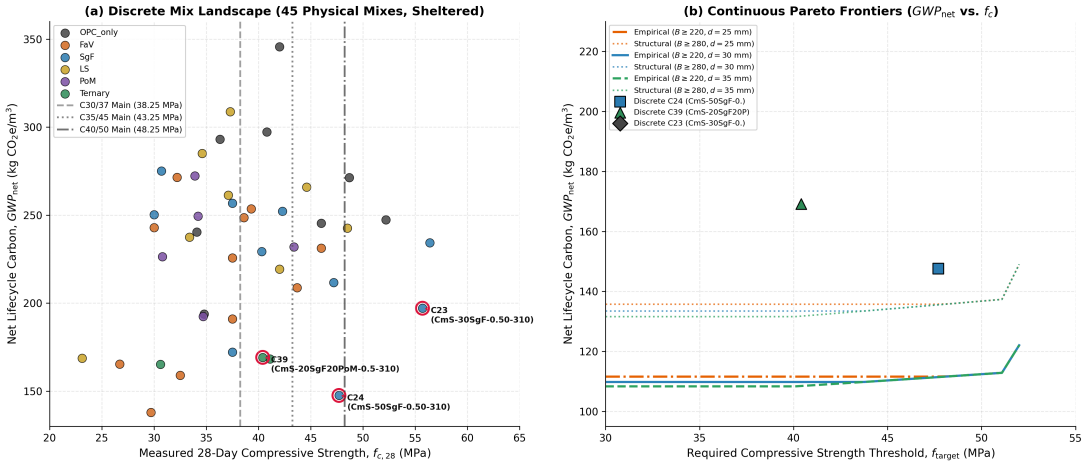


Fig. 3. Multi-objective Pareto frontiers: (a) Discrete mix landscape with buffered strength benchmarks; (b) Continuous Pareto trade-off frontiers across covers $(25, 30, 35$ mm) contrasting unconstrained envelope ($B \geq 220$ kg/m^3) against structural baseline ($B \geq 280$ kg/m^3, XC3).

TABLE VI
SYSTEMATIC SENSITIVITY AND ROBUSTNESS MATRIX ACROSS EIGHT OPTIMIZATION DIMENSIONS FOR BENCHMARK STRUCTURAL SCENARIO (GRADE C35/45, 30 MM COVER, SHELTERED EXPOSURE)

Dimension	Scenario	Optimal Mixture (w/b, B, SCM%)	Net GWP (kg/m^3)	Active Constraint
Exposure Class*	Sheltered Baseline (XC3)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	Unsheltered Code (XC4)	0.45, 300 kg, 25.5% FA, 24.5% SG	144.2	Strength ($f_c = 43.3$ MPa)
SCM Allocation	Economic (Baseline)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	Cut-Off (Waste)	0.51, 280 kg, 0.0% FA, 50.0% SG	129.4	Strength ($f_c = 43.3$ MPa)
	Mass Allocation	0.45, 280 kg, 24.0% FA, 26.0% SG	138.8	Durability ($d_{50} = 30.0$ mm)
Kinetic Model	Fickian ($n = 0.50$)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	Power-Law ($n = 0.509$)	0.45, 280 kg, 23.0% FA, 27.0% SG	133.5	Durability ($d_{50} = 30.0$ mm)
	Power-Law ($n = 0.452$)	0.45, 280 kg, 25.5% FA, 24.5% SG	133.9	Strength ($f_c = 43.3$ MPa)
CO_2 **Uptake**	Low Uptake	0.45, 280 kg, 24.0% FA, 26.0% SG	135.8	Durability ($d_{50} = 30.0$ mm)
	Midpoint Uptake	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	High Uptake	0.45, 280 kg, 24.0% FA, 26.0% SG	131.5	Durability ($d_{50} = 30.0$ mm)
Compaction Air	Air = 1.5%	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	Air = 2.0% (Baseline)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	Air = 3.0%	0.45, 280 kg, 12.1% FA, 37.9% SG	135.5	Strength ($f_c = 43.3$ MPa)
SCM Rules	Empirical (SCM $\leq 50\%$)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	Code (FA $\leq 35\%$)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
Strength Margin	IS 10262 Table 2 ($s = 5$, +8.25 MPa, Main)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability (0.48 MPa slack)
	Sensitivities: $s = 4$ (+6.6), LOO-CV (+4.4), Nom (0)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability (slacks 2.13, 4.33, 8.73 MPa)
Codified Baseline	EN 206 Baseline ($B \geq 280$ kg/m^3)	0.45, 280 kg, 24.0% FA, 26.0% SG	133.4	Durability ($d_{50} = 30.0$ mm)
	IS 456 Mild ($B \geq 300$ kg/m^3)	0.45, 300 kg, 24.0% FA, 26.0% SG	141.3	Durability ($d_{50} = 30.0$ mm)
	IS 456 Severe ($B \geq 320$ kg/m^3)	0.45, 320 kg, 24.0% FA, 26.0% SG	149.2	Durability ($d_{50} = 30.0$ mm)

*Outdoor difference (+10.8 kg/m^3) reflects codified binder ($B = 300$ vs 280 kg/m^3) and kinetics. IS 456 Mild ($B \geq 300$ kg/m^3) yields 141.3 kg/m^3 (6.3 kg/m^3 saving over C24); Severe ($B \geq 320$ kg/m^3) gives 149.2 kg/m^3 (C24 excluded). At 1.645σ (7.20 MPa, +26.1% kinetic), C24 reaches 31.6 mm $>$ 30 mm (fails); optimum (136.3 kg/m^3) saves 60.8 kg/m^3 (30.8%) over survivor C23 (197.1 kg/m^3). 1.0% PCE gives Net GWP 137.6–138.7 kg/m^3 (8.8–9.9 kg/m^3 headline saving over un-admixed C24; 14.6 kg/m^3 hypothetical like-for-like).

1) **Tight Cover (25 mm, Sheltered)**: The nominal optimum gives $f_c = 48.0$ MPa and $d_{50} = 25.0$ mm, saving 19.8% over nominal C39. Under +15% uncertainty ($d_{50} \leq 21.74$ mm), $f_c = 51.4$ MPa and $d_{50} = 21.7$ mm; saving over buffered C23 is 30.2% (27.5%–28.1% with 1.0% PCE).
2) **Standard Cover (30 mm, Sheltered)**: Nominal predictions are $f_c = 43.7$ MPa and $d_{50} = 30.0$ mm; under uncertainty ($d_{50} \leq 26.09$ mm), they become $f_c = 47.0$ MPa and $d_{50} = 26.1$ mm.
3) **Deep Cover (35 mm, Sheltered)**: The nominal capped mixture reaches $d_{50} = 34.8$ mm; unconstrained 35.3% FA + 14.7% SG reaches 35.0 mm with identical GWP. Under uncertainty, $d_{50} = 30.4$ mm.
4) **High Strength (30 mm, C40/50)**: The codified 48.25 MPa requirement selects the mixture in Table V ($f_c = 48.2$ MPa, $d_{50} = 24.8$ mm); +4.4 MPa LOO-CV instead gives 22.0% FA + 28.0% SG (133.8 kg/m^3).

D. Closed-Form Durability Limits and Kuhn-Tucker Optimality

For sheltered C35/45 at 30 mm cover, nominal (35.0 MPa), LOO-CV (39.4 MPa), and codified (43.25 MPa) thresholds select the same mixture in Table V, whose predicted strength is 43.73 MPa. The propositions explain this invariance on the fixed economic-allocation FA–SG branch; derivations appear in Appendix B-A.

Proposition 1 (Closed-Form Durability Limit): At $B = 280\ \mathrm{kg/m^3}$, $w/b = 0.45$, and $p_{\mathrm{SG}} = 50 - p_{\mathrm{FA}}$, sheltered kinetics reduce to $\ln k_{\mathrm{sh}} = 1.1172 + 0.01364 p_{\mathrm{FA}}$. Thus, $d_{50} \le d_{\mathrm{allow}}$ imposes

$$p_{\mathrm{FA}} \le p_{\mathrm{dur}} = \frac{\ln(d_{\mathrm{allow}}/\sqrt{50}) - 1.1172}{0.01364}. \tag{33}$$

Strength and the PPC fly ash cap may impose tighter upper bounds; Appendix B-A evaluates the benchmark.

Proposition 2 (Conditional Boundary Optimality): If the full lifecycle objective decreases throughout the feasible interval on this branch, its minimum occurs at the largest admissible FA content:

$$\frac{\mathrm{d}\mathcal{J}}{\mathrm{d}p_{\mathrm{FA}}} < 0 \implies p^*_{\mathrm{FA}} = \min\{50, p_{\mathrm{dur}}, p_{\mathrm{cap}}, p_{\mathrm{str}}\}, \tag{34}$$

provided the interval is nonempty; $p_{\mathrm{cap}} = 35\%$ and p_{str} are the PPC and strength bounds. Numerical derivatives support decreasing carbon on this economic-allocation branch, where benchmark durability binds and strength and FA caps remain slack. This conditional, fixed-branch result is not a global optimality proof. Cut-Off allocation instead favors $w/b = 0.53$ and 0% FA because low slag emissions and greater uptake alter the trade-off.

Local Cover Dividend: With durability in millimetres, the local 30 mm shadow price is $\mu^*_{\mathrm{dur}} = 0.4087 \approx 0.41\ \mathrm{kg\ CO_2e/(m^3 \cdot mm)}$, while $\mu^*_{\mathrm{str}} = 0$. For C35/45 under +8.25 MPa, strength binds before 35 mm (25.5% FA, $133.2\ \mathrm{kg/m^3}$). The durability-only 25–35 mm dividend therefore does not describe that full path (Table V).

Strength Slack and Compaction Limit: Codified strength slack is 0.48 MPa, exhausted at 2.11% air (LOO-CV: 2.99%). Batch air (5.1%) incurs a 13.5 MPa penalty and fails all three thresholds. Appendix B-A retains the baseline-air calculation.

VI. Discussion: Pareto Dynamics, Sensitivity, and Structural Caveats

A. Pareto Dynamics and Sensitivity Sweeps

Figure 3 synthesizes the structural Pareto landscape:

1) **Discrete Binder Clusters (Fig. 3a)**: Pure OPC clusters at 204–354 kg $\mathrm{CO_2e/m^3}$; fly ash reaches 140–180 $\mathrm{kg/m^3}$ but combines strength loss with $d_{50} = 45$–62 mm. Slag mixes (30%–50% SG) dominate through moderate strength loss and buffering (Table III).
2) **Cover Depth Carbon Dividend (Fig. 3b)**: Increasing cover from 25 to 35 mm lowers the frontier by $4.1\ \mathrm{kg\ CO_2e/m^3}$ through the FA-for-SG substitution quantified in Section V-D and Appendix B-A.
3) **Structural Binder Penalty**: Enforcing EN 206 XC3 binder ($B \ge 280\ \mathrm{kg/m^3}$) adds $+23.3$–$24.1\ \mathrm{kg\ CO_2e/m^3}$ ($+21.5\%$; benchmark $+23.6\ \mathrm{kg/m^3}$) over $B = 220\ \mathrm{kg/m^3}$ to ensure adequate paste lubrication.
4) **Optimization Gap Decomposition**: Continuous frontiers lie 14–60 $\mathrm{kg/m^3}$ below discrete batches. At sheltered C35/45, 30 mm cover, C24 ($147.57\ \mathrm{kg/m^3}$) versus the continuous design ($133.45\ \mathrm{kg/m^3}$) gives model-predicted $\Delta GWP_{\mathrm{total}} = \mathbf{14.1}\ \mathrm{kg\ CO_2e/m^3}$, pending batch trials. Decomposition gives: Path 1 (Binder→SCM) $10.0 + 4.1 = \mathbf{14.1}\ \mathrm{kg/m^3}$; Path 2 (SCM→Binder) $1.8 + 12.3 = \mathbf{14.1}\ \mathrm{kg/m^3}$. Binder reduction contributes $\approx 79\%$ and SCM blending $\approx 21\%$. At 25 mm cover, predicted savings over buffered C23 are $61.4\ \mathrm{kg/m^3}$ (31.2%). The PCE-qualified comparison is retained in Table VI.
5) **Corner and Boundary Extrapolation**: At the benchmark bounds described in Section V-D, nearest mixes in $(w/b, \mathrm{air}, \mathrm{SCM})$ space are: C28 ($d = 1.09$: $w/b = 0.50$, air 1.4%, 20% FA + 20% SG; f_c 41.1 vs. 41.7 MPa; k 4.374 vs. 4.381; d_{50} 30.9 vs. 31.0 mm), C23 ($d = 2.31$: $w/b = 0.50$, air 1.4%, 30% SG; f_c 55.7 vs. 49.2 MPa; k 2.534 vs. 2.970; d_{50} 17.9 vs. 21.0 mm), and C39 ($d = 2.33$: $w/b = 0.50$, air 1.6%, 20% SG + 20% PO; f_c 40.4 vs. 37.5 MPa; k 3.193 vs. 4.408; d_{50} 22.6 vs. 31.2 mm). Corner leverage $h_{00} = 0.250$ gives $s_{\mathrm{pred}} = s_e\sqrt{1 + 0.250} = 4.60$ MPa.

Figure 4 details the sensitivity sweep for benchmark C35/45 ($d_{\mathrm{cover}} = 30$ mm):

1) **Allocation (Fig. 4a)**: Mix rankings are invariant ($\rho > 0.997, p < 10^{-40}$) across regimes; clinker ($0.840\ \mathrm{kg/kg}$) dominates SCMs (14–56× economic: $\mathrm{EF_{FA}} = 0.015, \mathrm{EF_{SG}} = 0.060\ \mathrm{kg/kg}$; 7.0× mass: $0.030, 0.120$; cut-off: $0.005, 0.020$).
2) **Kinetics (Fig. 4b)**: The sheltered power-law ($n = 0.509$) gives a 50-year depth 28.91 mm vs. 26.40 mm Fickian ($+9.5\%$, mean $\bar{n} = 0.522$). The 10-year anchor shifts depth $+1.5\%$ (30.44 vs. 30.00 mm), within $+15\%$ margin ($d_{50,\mathrm{allow}} = 26.1$ mm). Outdoor $n = 0.452$ gives $d_{50} = 30.0$ mm (29.7% FA + 20.3% SG).
3) $\mathrm{CO_2}$ **Uptake (Fig. 4b)**: The uptake sweep shifts benchmark Net GWP by $+2.4/-1.9\ \mathrm{kg/m^3}$; offset ranges are given in Section III-D and Table II. High fly ash adds $5.8\ \mathrm{kg/m^3}$ uptake, but carbonation reaches 55 mm (Mix C35), depassivating 30 mm cover in 14.8 yr. Clinker reduction dominates uptake.
4) **Compaction (Fig. 4c)**: An air increase of $+1.0\%$ (3.0%) incurs a -4.35 MPa penalty, binding strength in Table VI. Section V-D gives the benchmark air-content failure thresholds.
5) **Fly Ash Cap (Fig. 4d)**: The PPC cap (FA $\le 35\%$, IS 1489 [17]) does not bind indoors (24.0% FA); at 35 mm, unconstrained FA reaches 35.3% (Table V), binding weakly (35.0% FA, 15.0% SG, $131.6\ \mathrm{kg/m^3}$). In XC4, the cap binds: C35/45 at 30 mm yields $142.3\ \mathrm{kg/m^3}$ (35.0% FA, 12.9% SG, 2.1% LS); at 25 mm, carbon is $142.2 \to 142.3\ \mathrm{kg/m^3}$; and C30/37 at 30 mm shifts $140.0 \to 141.0\ \mathrm{kg/m^3}$.

B. Structural Baselines and Practical Implementation

Optima at $B = 220\ \mathrm{kg/m^3}$ present two challenges: (1) **Paste Starvation**: neat paste volume is $176.6\ \mathrm{L/m^3}$ ($196.6\ \mathrm{L/m^3}$ with 2% air), below the 220–250 $\mathrm{L/m^3}$ limit [16]; codified baselines ($B \ge 280\ \mathrm{kg/m^3}$ sheltered, $B \ge 300\ \mathrm{kg/m^3}$

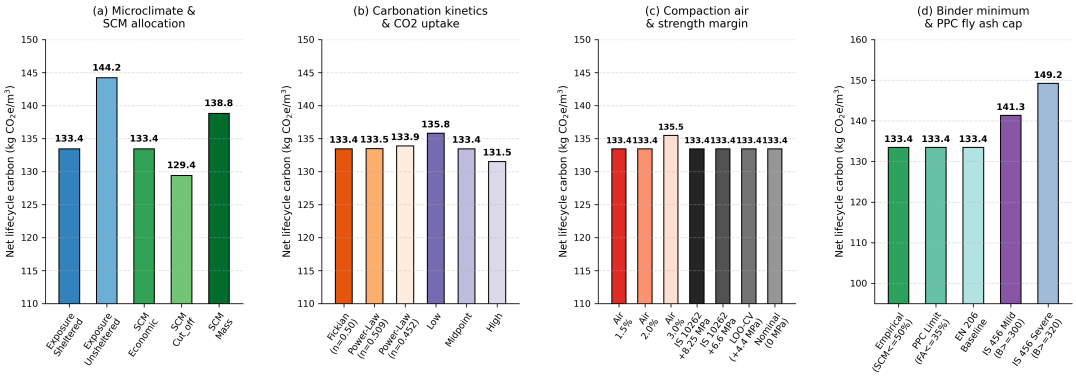


Fig. 4. Sensitivity sweep for benchmark C35/45 (30 mm cover): (a) Allocation ($\rho > 0.997$); (b) Kinetics and uptake; (c) Air and strength buffer; and (d) Binder minimum and PPC cap.

outdoor) restore paste to $224.8\ \mathrm{L/m^3}$. (2) **Clinker Depletion**: at 50% SCM, clinker falls to $110\ \mathrm{kg/m^3}$ ($\approx$ 15–20 $\mathrm{kg/m^3}$ portlandite reserve [9], [18]). Outdoor retardation (26%–30%) reflects pore-water blocking ($D_{\mathrm{aq}} \approx 10^{-4} D_{\mathrm{gas}}$). Storage effects were not statistically significant ($p > 0.50$). Slag mixes require 14–28 d moist curing. Regional kiln factors (0.893–0.940 kg/kg [4]) scale savings proportionally.

C. Limitations

Six physical and jurisdictional limitations qualify these findings: (a) Phenolphthalein indicates pH 9–9.5, whereas steel depassivates near pH 10.5–11.5; carbonation fronts may thus precede color change. (b) IS 456 Table 5 rates sheltered concrete as Mild ($B \geq 300\ \mathrm{kg/m^3}, w/b \leq 0.55$), where continuous optimum is $141.3\ \mathrm{kg/m^3}$ ($6.3\ \mathrm{kg/m^3}$ saving over C24; 1.8–2.9 $\mathrm{kg/m^3}$ with PCE), and coastal exposure as Severe ($B \geq 320\ \mathrm{kg/m^3}, w/b \leq 0.45$), where C24 is excluded and continuous optimum is $149.2\ \mathrm{kg/m^3}$. Chloride ingress is not modelled. (c) Setting $d_{50} = d_{\mathrm{cover}}$ leaves zero execution tolerance (5–10 mm); site cover must therefore be thicker. (d) Curing under 7 d increases carbonation by 25%–50% [19], exceeding the +15% margin. (e) Strength screening uses IS 10262 Table 2 (+8.25 MPa for $s = 5.0$ MPa, leaving 0.48 MPa slack at benchmark); corner error $s_{\mathrm{pred}} = 4.60$ MPa is within 4.13 MPa of nominal slack. The kinetic margin uses in-sample RMSE ($0.1411 \rightarrow +15\%$). The 1.645σ sensitivity in Table VI disqualifies C24 and compares the continuous design with feasible survivor C23. (f) The PCE compaction route remains uncalibrated on the physical cohort (Appendix A).

Low-Binder Extrapolation Envelope: Relaxing B to $220\ \mathrm{kg/m^3}$ yields theoretical Net GWP of 108.3–111.6 kg $\mathrm{CO_2e/m^3}$ indoors and 108.7–112.4 $\mathrm{kg/m^3}$ outdoors ($\sim 17.5\%$ reduction). The domain restriction in Section II-A and paste starvation above make these theoretical bounds rather than validated structural specifications.

VII. Conclusions and Structural Specification Guidance

Field calibration across 11 years in Chennai supports the discrete screening and continuous design framework. Under EN 206, Mix C24 dominates sheltered cases at ≥ 30 mm cover and outdoor C30/37–C35/45; IS 456 Severe excludes it, and the codified C40/50 strength margin selects C23. At 25 mm sheltered cover, kinetic uncertainty replaces nominal C39 with buffered C23 and increases carbon by 16.5% (Table V).

Across 18 scenarios, continuous optimization predicts 4.6%–31.2% carbon savings (1.1%–29.0% with 1.0% PCE). Binder reduction contributes most of the improvement; SCM blending provides additional savings subject to the strength, durability, and composition limits. The decomposition in Section VI-A remains model-predicted and requires batch trials. Codified binder baselines restore paste lubrication, while the low-binder envelope remains unvalidated.

Quality control remains decisive: strength binds at 3.0% air, and the batch-air penalty fails the targets; Section V-D distinguishes the codified and LOO-CV break-even limits. Reaching $\leq 2.0\%$ air requires PCE and 14–28 d moist curing. The PCE route is uncalibrated on the physical cohort.

Allocation-invariant rankings and the uptake offsets support prioritizing clinker displacement over carbonation uptake. This conclusion remains subject to the exposure, strength, cover, allocation, and uncertainty conditions stated above; neither the local analytical branch nor the extrapolated mixes establish global or experimentally validated optimality.

Disclosure of AI Assistance

ChatGPT (OpenAI) assisted with drafting parts of this research manuscript and refining its wording, clarity, and readability. The author retains responsibility for the manuscript's content and conclusions.

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APPENDIX A
STANDARDS AND TEST DETAILS

Codes, Grades & Protocols: Exposure follows EN 206 [14]; the grade mapping, IS 10262 margins, and IS 456 limits are specified in Sections III-B and IV. PPC fly ash follows IS 1489 [17]. Phenolphthalein (1%) testing followed RILEM CPC-18 / EN 12390-10 [11]; strength and air-content testing followed EN 12390-3 [20] and EN 12350-7 [21], respectively. Storage tests gave $p \geq 0.55$; the durations are reported in Section II-A. Module B1 follows EN 16757 (Annex BB) [13] and Lagerblad [15]. Achieving $\mathrm{Air} \leq 2.0\%$ at $w/b = 0.45$ requires PCE, adding $+4.2$–5.3 kg $\mathrm{CO_2e/m^3}$ [22]; the qualified carbon comparison is given in Table VI. No dataset mix used PCE, so this compaction route remains uncalibrated. Data: Mendeley Data [12] ([11]). Scripts: https://github.com/abhitejj/low-carbon-concrete-optimization. The study was conducted in *MESS 2026*, under Prof. R. N. Singh and TA Gayathri Vangala.

APPENDIX B
DETAILED MIXTURES, SCREENING COUNTS, AND ANALYTICAL DERIVATIONS

Table VII retains all 18 scenarios from Table V. Targets for C30/37, C35/45, and C40/50 are 30.0, 35.0, and 40.0 MPa. Continuous solutions use $w/b = 0.45$ and a +8.25 MPa strength margin. Constituent percentages are by binder mass; a dash denotes zero replacement. Carbon is net GWP in kg CO_2e/m^3.

TABLE VII
FULL CONTINUOUS COMPOSITIONS AND DISCRETE FEASIBILITY COUNTS

		Feasible discrete mixes			Continuous: baseline kinetics					Continuous: +15% kinetics				
Grade	**Cover (mm)**	**Nom.**	**Kin.**	**Joint**	**FA**	**SG**	**LS**	**PO**	**Carbon**	**FA**	**SG**	**LS**	**PO**	**Carbon**
					(% binder)					**(% binder)**				
A. Sheltered (XC3): 35 candidates														
C30/37	25	24/35	19/35	13	10.7	39.3	–	–	**135.7**	0.4	49.6	–	–	**137.5**
C30/37	30	30/35	26/35	18	24.0	26.0	–	–	**133.4**	13.8	36.2	–	–	**135.2**
C30/37	35	32/35	30/35	19	35.0	15.0	–	–	**131.6**	25.1	24.9	–	–	**133.3**
C35/45	25	20/35	16/35	9	10.7	39.3	–	–	**135.7**	0.4	49.6	–	–	**137.5**
C35/45	30	23/35	21/35	11	24.0	26.0	–	–	**133.4**	13.8	36.2	–	–	**135.2**
C35/45	35	25/35	23/35	12	25.5	24.5	–	–	**133.2**	25.1	24.9	–	–	**133.3**
C40/50	25	15/35	12/35	5	10.1	39.9	–	–	**135.8**	0.4	49.6	–	–	**137.5**
C40/50	30	17/35	16/35	5	10.1	39.9	–	–	**135.8**	10.1	39.9	–	–	**135.8**
C40/50	35	18/35	17/35	5	10.1	39.9	–	–	**135.8**	10.1	39.9	–	–	**135.8**
B. Unsheltered (XC4): 26 candidates														
C30/37	25	24/26	22/26	14	35.0	12.2	–	2.8	**142.2**	28.6	21.4	–	–	**143.7**
C30/37	30	25/26	24/26	14	35.0	9.7	5.3	–	**142.0**	35.0	10.4	2.5	2.1	**142.0**
C30/37	35	25/26	25/26	14	35.0	9.7	5.3	–	**142.0**	35.0	9.7	5.3	–	**142.0**
C35/45	25	17/26	16/26	9	25.5	24.5	–	–	**144.2**	25.5	24.5	–	–	**144.2**
C35/45	30	17/26	17/26	9	25.5	24.5	–	–	**144.2**	25.5	24.5	–	–	**144.2**
C35/45	35	17/26	17/26	9	25.5	24.5	–	–	**144.2**	25.5	24.5	–	–	**144.2**
C40/50	25	13/26	13/26	5	10.1	39.9	–	–	**146.9**	10.1	39.9	–	–	**146.9**
C40/50	30	13/26	13/26	5	10.1	39.9	–	–	**146.9**	10.1	39.9	–	–	**146.9**
C40/50	35	13/26	13/26	5	10.1	39.9	–	–	**146.9**	10.1	39.9	–	–	**146.9**

Common exposure conditions. Exposure bounds follow Section IV. "Nom." counts nominal-strength/nominal-kinetics survivors; "Kin." adds +15% kinetic uncertainty alone; "Joint" applies both strength and kinetic buffering. Counts follow the same grade/cover order as Table V.

Properties of selected discrete mixes. C39 has measured strength 40.4 MPa and sheltered $d_{50} = 22.6$ mm; C24 has measured strength 47.7 MPa, sheltered $d_{50} = 25.04$ mm, and unsheltered $d_{50} = 17.5$ mm.

Screening and admixture qualifications. Nominal and buffered screening follow Section III-B. C39 also fails kinetic uncertainty alone ($25.99 > 25$ mm). The original mixes contained no PCE; adding 1.0% PCE within the optimizer shifts benchmark Net GWP to 137.6–138.7 kg/m^3. The 14.6 kg/m^3 saving applies only to hypothetical identical PCE addition to C24, which requires none.

A. Derivations for Closed-Form Durability and Boundary Optimality

Durability reduction. Substitution into Eq. 12 gives

$$\begin{aligned}\ln k_{\mathrm{sh}} &= -1.2939 + 4.0681(0.45) + 0.02525p_{\mathrm{FA}} + 0.01161(50 - p_{\mathrm{FA}}) \\ &= 1.1172 + 0.01364p_{\mathrm{FA}}.\end{aligned}$$

Enforcing $\ln k_{\mathrm{sh}} \le \ln(d_{\mathrm{allow}}/\sqrt{50})$ proves Proposition 1. At 30 mm, $\ln(30/\sqrt{50}) = 1.4452$ gives 24.04% FA and 25.96% SG. Durability limits are 10.7% FA at 25 mm, 13.8% at 26.09 mm (+15% kinetics at 30 mm cover), and 35.3% at 35 mm, reduced to 35.0% by the PPC cap; strength may impose a tighter limit.

Objective slope and endpoint argument. Under economic allocation, the binder slope is $\frac{B}{100}(EF_{\mathrm{FA}} - EF_{\mathrm{SG}}) = -0.126$ kg CO_2e/(m$^3 \cdot$ %FA). With $\rho_{\mathrm{FA}} = 2300 < \rho_{\mathrm{SG}} = 2900$ kg/m^3, aggregate displacement gives an embodied slope of -0.130; the benchmark uptake slope is $+0.037$, giving $\mathrm{d}\mathcal{J}/\mathrm{d}p_{\mathrm{FA}} = -0.167$ in the same units. Derivatives at $p_{\mathrm{FA}} = 0, 10, 24.04, 40, 50\%$ are $-0.174, -0.172, -0.167, -0.158, -0.149$, respectively. These samples support monotonicity but do not prove its sign between samples. Conditional on a negative derivative throughout the feasible interval, increasing FA lowers $\mathcal{J}$ until durability, strength, or the composition cap binds, proving Proposition 2 on this fixed branch. Clinker emissions (0.840 kg/kg) exceed FA and SG (0.015 and 0.060); this economic-allocation ordering alone does not prove global optimality.

Cover dividend and compaction arithmetic. Where durability controls, $\mathrm{d}p^*_{\mathrm{FA}}/\mathrm{d}d_{\mathrm{allow}} = 1/(0.01364d_{\mathrm{allow}})$ and the physical carbon dividend is $-(\mathrm{d}\mathcal{J}/\mathrm{d}p_{\mathrm{FA}})/(0.01364d_{\mathrm{allow}})$. At 30 mm, $0.1672/(0.01364 \times 30) = 0.4087$ kg CO_2e/(m$^3 \cdot$ mm). KKT multiplier values depend on constraint scaling. The durability-controlled 25–35 mm branch gives $(135.7 - 131.6)/10 = 0.41$, with interval slopes 0.46 (25–30 mm) and 0.36 (30–35 mm). These branch values apply while strength remains slack, as for C30/37; they are not the complete codified C35/45 path, which reaches 133.2 kg/m^3 at 35 mm. Benchmark strength slacks are $43.73 - 43.25 = 0.48$ MPa codified, 4.33 MPa LOO-CV, and 8.73 MPa nominal. Break-even air is $2.0\% + 0.48/4.354 \approx 2.11\%$ codified and $2.99\% \approx 3.0\%$ LOO-CV.